

Coding & Solution

Date	5 July2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/GUTTI09/OptiCrop-Smart-Agricultural-Production-Optimization-Engine
Programming Language(s)	Python, HTML5, CSS3, JavaScript
Framework(s) Used	Flask,scikit-learn.Bootstrap
Key Features Implemented	• User Registration & Login • AI-based Crop Prediction • Soil Data Input (N, P, K, pH, Temperature, Humidity, Rainfall) • Crop Recommendation Results • Responsive Web Interface
Pending / Incomplete Features	• Weather API Integration (Optional) • Market Price Prediction • Fertilizer Recommendation Module
Setup / Run Instructions	1. Install Python and required libraries (Flask, scikit-learn, pandas, numpy, joblib). 2. Clone the GitHub repository. 3. Open the project in Visual Studio Code or PyCharm. 4. Run app.py. 5. Open http://127.0.0.1:5000 in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The application follows a modular architecture with separate frontend, backend, machine learning model, and database components. It successfully predicts suitable crops based on soil and environmental parameters and provides an intuitive interface for users.